

NUTRITION (CELEBRATIONS)

Imagine driving a brand new and expensive car and after a few hours of pure enjoyment you notice that the fuel gauge is showing close to empty, so you pull into the first gas station you find. Looking in the owner's manual you see that the manufacturer has recommended only premium fuel for the high-compression engine. You decide, however, that any grade fuel will do and fill it with "regular" instead. After all, you think, they look and smell the same. Later, when you check the engine oil, you top it up with a little water. How long do you think your dream car will last?

Our bodies are far more beautiful and complex than the finest machine ever made by humans. Like a fine sports car, our bodies need fuel to power our lives, and that **fuel comes from the food we eat**. A balanced diet chosen from the best **foods will provide the essential nutrients needed for growth, maintenance, and energy**. If we choose low-quality foods or do not eat enough of even the best foods, the body machinery will suffer. **Too much food may result in obesity, and excessive amounts of some nutrients may cause toxicity.**



The right fuel :

Let us celebrate every meal with healthful food choices. In each meal one can enjoy the abundance of nutrient-dense, whole-grain breads and cereals, along with rich, **colorful** fruits and vegetables containing abundant amounts of vitamins, minerals, dietary fibers, and phytochemicals. Enjoy essential fats in crunchy nuts and seeds, bone-building calcium from low-fat milk or a fortified soymilk, and healthful proteins from satisfying legumes (beans, peas, and lentils) seasoned delicately with herbs and small amounts of salt, sugar, and vegetable oil as needed. These energizing, body-building foods consumed daily in appropriate quantities can reduce the risk of cancer, coronary heart disease, hypertension, intestinal disease, obesity, and osteoporosis. We can truly celebrate at every meal because of the abundance of good food God has given us!

Understanding nutrition As we feed ourselves physically, we'd likewise feed ourselves spiritually. How can we feast on God's Word? As we feed ourselves physically, we'd likewise feed ourselves spiritually. How can we feast on God's Word?

Many people think that **choosing** good nutrition is difficult, and **understanding** it even harder. Both tasks, however, are really very simple.

Let's start with the "understanding" part.

Our bodies get the nutrients we need from the food we eat as the food is digested and assimilated in a fantastic process that begins in the mouth, moves to the stomach, then to the small intestines, and finally to the large bowel.

The nutrients our bodies need include:

Carbohydrates (Glucins): They are the main source of energy for all functions of the body, and provide calories quickly. The largest portion of our diet should come from these in as unrefined a form as possible. Whole grains, legumes, fruits, and vegetables are rich in these. There are approximately 4 kcal (a unit used to express the amount of energy contained in food) per gram in carbohydrates. Apart from sugars like monosaccharides (glucose, fructose) and disaccharides (saccharose, maltose, lactose), we also have polysaccharides that mainly found in cereal grains (wheat, rice, corn...) and in roots and tubers (potatoes, cassava,...).

They are called carbohydrates because of the carbon, hydrogen and oxygen they contain, are usually in the proportion to form water. The general formula $C_n(H_2O)_n$. C:Carbon, H:Hydrogen, O:oxygen and n stands for a number varying with the type of the Carbohydrates. According to the WHO, in an average diet of 2000 calories is required by the body daily.

Proteins: Every cell in the body contains proteins. They form the basis of the body structure, they are constituents of muscles, blood, skin and internal organs. Bones are formed by collagen proteins upon which calcium and other minerals are based. 17% of our weight is made by proteins.

Proteins are **not stored** in the body like carbs or fats, that's why they have to be ingested throughout life.

Tissue repair and growth require them. While almost all foods contain some protein, particularly good sources are milk, eggs, and other animal products. Légumes (beans, lentils, peas...) are excellent plant sources. Each gram of protein yields 4 kcal (like the carbohydrates).

Proteins are made by amino acids, and they differ by the kind of amino acids that constitute them and in which order.

Our body needs amino acids not specific proteins. All experts agree, and experience has demonstrated that **meat is not indispensable in human diet for the enjoyment of good health**. In fact, diet high in meat and other animal and dairy products provides too much proteins and animal fats, that increase the risk of atherosclerosis (blood vessels narrowing and plaque formation within them) leading to cardiovascular diseases and other NCDs (Non communicable disease) such as hypertension, heart diseases...

Large quantities of proteins can also cause kidney illnesses, excessive wastes of toxins in the body like cadaverine (from animal products like meat), putrescine, uric acid, indole,...(35 different kind of toxins that are harmful to the body).

Pregnant women (60 gr/day) and lactating women (65 gr/day) need more proteins . Normally the body needs 52.5 gr/day for a man weighing 70 kg (US Food Nutrition Board). People who have enough food tend to consume more proteins than needed.

Fats: They are all insoluble in water and are a concentrated source of energy. We often get too much fat in our diet because we like the flavor it imparts to foods (e.g., boiled or baked potatoes versus French fries).

Lard, shortening, butter, fish, meat, poultry, margarine, fried foods, cream, mayonnaise, and cheese are and cheese are not needed and contain harmful fats.

Avocado, olives, seeds, and nuts contain healthy fats. Omega 3 fats are very healthy – found in walnuts and ground flax seeds.

Vitamins, minerals and trace of minerals

Vitamins (fat soluble and water soluble) are essential for growth, they synthesise other nutrients, keep us mentally strong, agile and contribute to the health of many organs and reinforce immune defences.

They are antioxidants and phytochemicals and protect the body from diseases and some of the effects of aging. They are found primarily in fruits and vegetables. They are also found in whole grain cereals, seeds and nuts...

We can get all the needed vitamins and minerals from natural foods. Nevertheless when nutrients start being concentrated (in tablets, food supplements..), that's the beginning of the problems with the loss of the natural balance.

Water

Water is life. Some people argue on the fact that water is not a nutrient. Nevertheless if we consider the definition of a nutrient « a substance or an ingredient that promote growth, provides energy and maintains life », and knowing that water sustains life, we can unmistakably consider it as a nutrient.

Our body needs a certain amount of water every day. To know the required amount, take your weight (in kg)/30, you find the amount of water your body needs daily. Ex: if you have 60kg, you need at least 2 liters of water daily.

Note that when you exercise, when it is hot, when you've done some physical activity...you have to increase the amount of water consumed. It must be taken the whole day, not once.

We can get all the needed vitamins and minerals from natural foods. Nevertheless when nutrients start being concentrated (in tablets, food supplements..), **that's the beginning of the problems with the loss of the natural balance**. Let us eat natural and fresh foods, they are the right fuel.

Essential food groups

These groups may be divided into 5 categories where we can find all the needed ingredients :

1. **Cereals and grains** : Cereals and Grains: These should form the **foundation of our diet**. They include whole-grain breads, pastas, rice, and corn. They are rich in dietary fiber and complex carbohydrates, as well as an array of vitamins and minerals when taken from **unrefined** (not white) sources. Depending on a person's age, weight, and activity levels, 6 to 12 servings from this group should be consumed each day.



Cereals and grains

Fruits and vegetables

Légumes, nuts, seeds

Dairy and eggs

2. **Fruits and vegetables** :These foods come in a wide variety of colors, flavors, and textures and are the richest sources of protective phytochemicals, antioxidants, and many vitamins and minerals. Depending on body size, age, and activity levels, at least 5 to 10 servings of these foods chosen from a range of colors should be consumed daily. Many people seem to prefer fruits over vegetables, but we need a balance of both. Foods in this group that are the deepest in color often have the largest amounts of phytochemicals and antioxidants. Fruit juices should be limited to no more than 1 small serving per day.
3. **Legumes, nuts and seeds** : Legumes such as beans, peas, and lentils are an important source of good protein, along with minerals, vitamins, and other protective elements. Three to 5 servings of these should be included in the daily diet, depending on a person's age and weight. Nuts and seeds are excellent sources of essential fats, but because they are a concentrated source of calories, they should be limited to no more than 1 to 2 servings per day. Non-vegetarians would include fish, fowl, and meat in this group, but should consume only moderate amounts.
4. **Dairy and Eggs** (or fortified equivalents): These animal sources of food provide many important nutrients, including calcium and vitamin B₁₂. Vitamin B₁₂ is found only in animal products and prevents pernicious anemia and neurological disorders. It also promotes normal cellular division. Individuals who choose not to consume any animal products need to eat sufficient foods fortified with vitamin B₁₂ or take it in supplement form. It's very important to read the food labels of the equivalent foods to make certain that they are adequately fortified. Deficiency symptoms of vitamin B₁₂ can take 4 to 6 years to develop after all intake has been stopped. By the time problems have been discovered, permanent damage may already have occurred.
5. **Fats, Oils, Sweets, and Salt**: These foods are required only **in small amounts**. The essential fats and sodium are necessary for optimum health. Iodine is a necessary trace mineral and is easily supplied if iodized salt is used; it also can be obtained from sea salt, seaweed, or a supplement. **Refined sugar is not required for good health**, but small amounts add palatability and flavor to our foods.

Make half your plate fruits and vegetables

One of the most important keys to eating a balanced plant-based diet is selecting a variety of foods whose color, texture, and flavor add interest to the diet. These foods are best when consumed as they come from nature: **not refined, not pulled apart, not fractionated**. Whole foods should be the goal.

“Use plant foods as the foundation of your meals. . . . Eating a variety of grains (especially whole-grains), fruits and vegetables is the basis of healthful eating.” This recommendation has been simplified in the 2010 “Dietary Guidelines for Americans” to “**Make half your plate fruits and vegetables.**”

Today the world is recognizing the **advantages** of a vegetarian diet:

- Low in fat, particularly saturated fat
- No cholesterol (with a total vegetarian diet)
- High in dietary fibers
- Low in refined sugar (need to avoid highly sweetened items, even if from plant foods)
- Contains rich sources of vitamins and minerals
- Contains high amounts of protective substances such as phytochemicals, antioxidants, etc.

Guiding principles of food choices

A healthful diet needs to be based on sound principles that guide the food choices we make. We would like to suggest five:

1. **Variety:** Perhaps the most important principle of eating right is selecting a variety of foods from the five groups discussed earlier in this chapter (cereals and grains; fruits and vegetables; legumes, nuts, and seeds; dairy and eggs or equivalents; and fats, oils, and salt). This **ensures a wide range of nutrients to support a healthy body, and the various textures, tastes, and colors enhance the pleasure of eating.**
2. **Quality:** Choose the majority of your food from **whole foods**—not refined foods. These foods are **nutrient-dense rather than calorie-dense.**



Variety



Quality



Moderation



Avoidance

3. **Balance:** Obesity is a growing problem worldwide. There needs to be a **balance between the amount of energy we eat** (foods) and the **energy we expend** (physical activity) if we are to maintain a healthy weight.
4. **Moderation:** Some important components of a healthful diet need to be eaten only in small amounts. These would include fats and salt. We require adequate amounts of the essential fats. Fats are also the vehicle for fat-soluble vitamins. We also need small amounts of salt to maintain our electrolytes.

5. Avoidance: **Highly refined foods** that often have large amounts of their nutritional elements removed **should be avoided**, as should foods and beverages that have no nutritional value (alcohol, coffee, and sodas).

Many excellent online tools are available that allow you to track and analyze what you eat every day. One of the best is SuperTracker, which is free for anyone to use.

The quality of life

A **healthful diet can increase lifespan and the quality of life**. God loves us and desires that we lead healthy, productive, and happy lives. We can celebrate His goodness as we appropriately enjoy the many products of the earth that He has given us.



Healthful Diet for longevity



Spiritual food

Spiritual food :

Just as we require physical food each day, we also need to feed the inner person on spiritual food. We should not neglect to make a daily practice of feeding on God's Word. We have emphasized the need for variety, balance, and flavor in the foods we consume, but we require balance in our spiritual food, as well. **We can feast on God's Word by contemplating His wonderful promises, reading inspirational stories and exhortations, and spending time daily in prayer.** These practices will help us to grow spiritually as well as physically. Balance and control in life **come from the steady application of the lessons learned in the reading of His Word.**

Let us do these things with praise in our hearts for the energy and health that God provides.

SELF-EVALUATION QUESTIONS

1. Which types of essential food groups do we have? Provide 3 examples of food known in Rwanda for each group.
2. Which different nutrients needed by our body do we find in food ?
3. What is the difference between nutrients and essential food ? Provide some example in each category.
4. Please provide the guiding principles of food choices for a healthful diet. How to define these principles ?
5. As we look for a balanced diet in food, we'd also make sure that are spritual food is balanced. How can you achieve that.
6. Why should we promote vegetarian diet ?